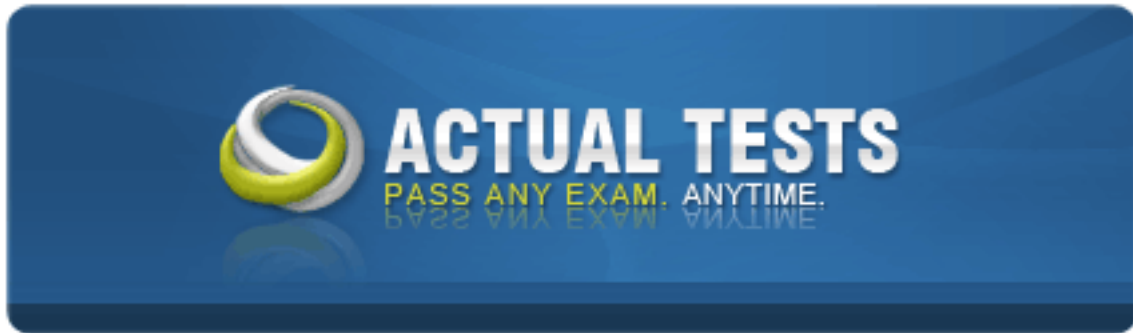


Cisco 640-878



**Building Cisco Service Provider Next-Generation
Networks, Part 2**
Version: 6.0

QUESTION NO: 1

Which spanning-tree mode can be used to map several VLANs to a single spanning-tree instance?

- A. MST
- B. PVST+
- C. PVRST+
- D. RSTP

Answer: A

Explanation:

QUESTION NO: 2

Which spanning-tree mode uses the name command to configure a region name?

- A. MST
- B. PVST+
- C. PVRST+
- D. RSTP

Answer: A

Explanation:

QUESTION NO: 3

Which procedure is used as the last resort disaster recovery procedure to completely replace the currently installed IOS XR software on Cisco IOS XR routers?

- A. netboot
- B. turboboot
- C. install recovery
- D. install rollback
- E. install add and install activate

Answer: B

Explanation:

QUESTION NO: 4

Which file extension indicates a bootable installation file in Cisco IOS XR software?

- A. .bin
- B. .tar
- C. .smu
- D. .pie
- E. .vm
- F. .mini

Answer: E

Explanation:

QUESTION NO: 5

When configuring an ACL entry, which network and wildcard mask matches only IP addresses 10.8.144.0 to 10.8.151.255?

- A. 10.8.144.0 0.0.3.255
- B. 10.8.144.0 0.0.7.255
- C. 10.8.144.0 0.0.15.255
- D. 10.8.144.0 0.0.252.255
- E. 10.8.144.0 0.0.248.255
- F. 10.8.144.0 0.0.240.255

Answer: B

Explanation:

QUESTION NO: 6

Which IP addresses are matched by the permit 192.168.80.64 0.0.0.15 access-list entry?

- A. 192.168.80.64 to 192.168.80.255
- B. 192.168.80.64 to 192.168.80.96
- C. 192.168.80.64 to 192.168.80.95
- D. 192.168.80.64 to 192.168.80.80
- E. 192.168.80.64 to 192.168.80.79

Answer: E

Explanation:**QUESTION NO: 7**

Which two statements about NAT444 are true? (Choose two.)

- A. NAT444 packets are translated three times.
- B. NAT packets traverse three IPv4 addressing domains.
- C. NAT444 needs to work together with DNS64.
- D. There are two types of NAT444 (stateful or stateless).
- E. NAT is performed by the CPE and also by the service provider router.

Answer: B,E

Explanation:**QUESTION NO: 8**

Refer to the exhibit.

```
ipv4 access-list FILTER
10 permit tcp any 192.168.15.32 0.0.0.15 eq www
20 deny ipv4 any 192.168.15.32 0.0.0.15
30 permit ipv4 any any
```

The access list has been configured on the Gi0/0/0/0 interface in the inbound direction. Which four packets that are sourced from 10.1.1.1 TCP port 1060, if they are routed to the Gi0/0/0/0 interface, will be permitted? (Choose four)

- A. destination IP address: 192.168.15.37, destination TCP port: 8080
- B. destination IP address: 192.168.15.41, destination TCP port: 8080
- C. destination IP address: 192.168.15.49, destination TCP port: 8080
- D. destination IP address: 192.168.15.36, destination TCP port: 80
- E. destination IP address: 192.168.15.46, destination TCP port: 80
- F. destination IP address: 192.168.15.49, destination TCP port: 80

Answer: C,D,E,F

Explanation:

QUESTION NO: 9

Which two statements about NAT64 are true? (Choose two.)

- A. NAT64 packets traverse two IPv4 addressing domains.
- B. NAT64 packets are translated two times.
- C. There are two types of NAT64 (stateful or stateless).
- D. NAT is performed by the CPE and also by the service provider edge router.
- E. The DNS64 server embeds the IPv4 address from the DNS A record with a preconfigured IPv6 translation prefix.

Answer: C,E

Explanation:

QUESTION NO: 10

Refer to the exhibit.

```
ipv4 access-list FILTER
10 deny tcp any 10.10.192.0 0.0.3.255 eq telnet
20 permit ipv4 any 10.10.192.0 0.0.3.255
30 deny ipv4 any any
```

The access list has been configured on the Gi0/0/0/0 interface in the inbound direction. Which two packets that are sourced from 172.16.1.1 TCP port 1050, if they are routed to the Gi0/0/0/0 interface, will be permitted? (Choose two)

- A. destination IP address: 10.10.192.201, destination TCP port: 80
- B. destination IP address: 10.10.193.255, destination TCP port: 80
- C. destination IP address: 10.10.196.1, destination TCP port: 80
- D. destination IP address: 10.10.195.254, destination TCP port: 23
- E. destination IP address: 10.10.193.145, destination TCP port: 23
- F. destination IP address: 10.10.197.32, destination TCP port: 23

Answer: A,B

Explanation:

QUESTION NO: 11

Which first-hop router redundancy protocol uses the active virtual gateway to assign a virtual MAC address to the active virtual forwarders?

- A. HSRP
- B. VRRP
- C. GLBP
- D. FHRP

Answer: C

Explanation:

QUESTION NO: 12

Which two statements about the BGP weight attribute on Cisco routers are true? (Choose two.)

- A. It is a well-known discretionary attribute.
- B. It is only locally significant.
- C. Routes with higher weight are the preferred routes.
- D. It is used to influence the inbound traffic from the upstream AS.
- E. It is set to 100 by default.

Answer: B,C

Explanation:

QUESTION NO: 13

Which BGP attribute is also used for loop prevention?

- A. weight
- B. local preference
- C. MED
- D. AS path

Answer: D

Explanation:

QUESTION NO: 14

The permit 172.16.32.0 0.0.15.255 access-list entry permits exactly which IP address range?

- A. 172.16.32.0 to 172.16.47.255
- B. 172.16.32.0 to 172.16.48.255
- C. 172.16.32.0 to 172.16.62.255
- D. 172.16.32.0 to 172.16.63.255
- E. 172.16.32.0 to 172.16.64.255
- F. 172.16.32.0 to 172.16.255.255

Answer: A

Explanation:

QUESTION NO: 15

On the Cisco ME 3400 switch, spanning tree is enabled by default on which type of switch port?

- A. UNI
- B. ENI
- C. NNI
- D. ENI and NNI
- E. UNI, ENI, and NNI

Answer: C

Explanation:

QUESTION NO: 16

Refer to the two show output examples below. The switch with the e8ba.70b5.7180 MAC address is the root bridge for which VLAN or VLANs?

S78#show spanning-tree

MST0

Spanning tree enabled protocol mstp

Root ID Priority 32768

Address e8ba.70b5.6c00

Cost 0

Port 23 (FastEthernet0/21)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32768 (priority 32768 sys-id-ext 0)

Address e8ba.70e1.d980

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	200000	128.3	P2p
Fa0/2	Desg	FWD	200000	128.4	P2p
Fa0/21	Root	FWD	200000	128.23	P2p
Fa0/23	Altn	BLK	200000	128.25	P2p

MST1

Spanning tree enabled protocol mstp

Root ID Priority 24577

Address e8ba.70b5.6c00

Cost 200000

Port 23 (FastEthernet0/21)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address e8ba.70e1.d980

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	200000	128.3	P2p
Fa0/21	Root	FWD	200000	128.23	P2p
Fa0/23	Altn	BLK	200000	128.25	P2p

MST2

Spanning tree enabled protocol mstp

Root ID Priority 24578

Address e8ba.70b5.7180

Cost 200000

Port 25 (FastEthernet0/23)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32770 (priority 32768 sys-id-ext 2)

Address e8ba.70e1.d980

Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/2	Desg FWD	200000	128.4	P2p	
Fa0/21	Altn BLK	200000	128.23	P2p	
Fa0/23	Root FWD	200000	128.25	P2p	

S78# show spanning-tree mst

```
##### MST0  vlans mapped: 1-69,71-79,81-4094
Bridge      address e8ba.70e1.d980 priority 32768 (32768 sysid 0)
Root        address e8ba.70b5.6c00 priority 32768 (32768 sysid 0)
            port Fa0/21      path cost 0
Regional Root address e8ba.70b5.6c00 priority 32768 (32768 sysid 0)
            internal cost 200000 rem hops 19
Operational hello time 2, forward delay 15, max age 20, txholdcount 6
Configured  hello time 2, forward delay 15, max age 20, max hops 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/1	Desg FWD	200000	128.3	P2p	
Fa0/2	Desg FWD	200000	128.4	P2p	
Fa0/21	Root FWD	200000	128.23	P2p	
Fa0/23	Altn BLK	200000	128.25	P2p	

```
##### MST1  vlans mapped: 70
Bridge      address e8ba.70e1.d980 priority 32769 (32768 sysid 1)
Root        address e8ba.70b5.6c00 priority 24577 (24576 sysid 1)
            port Fa0/21      cost 200000 rem hops 19
```

Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/1	Desg FWD	200000	128.3	P2p	
Fa0/21	Root FWD	200000	128.23	P2p	
Fa0/23	Altn BLK	200000	128.25	P2p	

```
##### MST2  vlans mapped: 80
Bridge      address e8ba.70e1.d980 priority 32770 (32768 sysid 2)
Root        address e8ba.70b5.7180 priority 24578 (24576 sysid 2)
            port Fa0/23      cost 200000 rem hops 19
```

Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/2	Desg FWD	200000	128.4	P2p	
Fa0/21	Altn BLK	200000	128.23	P2p	
Fa0/23	Root FWD	200000	128.25	P2p	

- A. VLAN 1
- B. VLAN 70

- C. VLAN 80
- D. VLANs 1, 70, and 80
- E. VLANs 1 and 70
- F. VLANs 1 and 80
- G. VLANs 70 and 80

Answer: C

Explanation:

QUESTION NO: 17

Refer to the Cisco IOS XR commands exhibit.

```
RP/0/RSP0/CPU0:PE7#install activate disk0:asr9k-video-p-4.1.0
                        ^
% Invalid input detected at '^' marker.

RP/0/RSP0/CPU0:PE7#show user all
Tue Sep 20 21:39:00.331 UTC
Username: admin
Groups: root-system
<output omitted>
```

The router administrator is trying to activate a software package on the router but is not able to do so. Which statement about this problem is true?

- A. The router needs to be in the global configuration mode.
- B. The router needs to be in the admin global configuration mode.
- C. The router needs to be in the admin EXEC mode.
- D. The install activate command is not the correct command to use.
- E. The administrator needs to log in as the "root" user instead of the "admin" user.

Answer: C

Explanation:

QUESTION NO: 18

During the BGP route selection process on Cisco routers, which BGP attribute is examined first, to determine the best path to use?

- A. AS path
- B. MED
- C. local preference
- D. weight
- E. origin

Answer: D

Explanation:

QUESTION NO: 19

Which type of service provider is responsible for offering backbone connectivity services to other service providers?

- A. CSP
- B. TSP
- C. NSP
- D. ISP
- E. ASP

Answer: C

Explanation:

QUESTION NO: 20

Cisco IP NGN service providers can offer multiple services to their customers by using which type of technology in their core networks?

- A. ATM
- B. SONET/SDH
- C. Frame Relay
- D. IP/MPLS
- E. PBB

Answer: D

Explanation:

QUESTION NO: 21

ISPs use which protocol to establish peering sessions in an Internet exchange point environment?

- A. LDP
- B. IS-IS
- C. BGP
- D. OSPF

Answer: C

Explanation:

QUESTION NO: 22

You are enabling OSPF on a router and notice that all the Fast Ethernet and the Gigabit Ethernet interfaces have the same OSPF cost of 1. Which single configuration change can you make in router ospf configuration mode so that the Fast Ethernet interfaces have a higher OSPF cost than the Gigabit Ethernet interfaces?

- A. Change the OSPF interface cost globally.
- B. Change the OSPF default metric.
- C. Change the OSPF auto-cost reference bandwidth.
- D. Change the OSPF administrative distance.
- E. Change the OSPF metric type from E2 to E1.

Answer: C

Explanation:

QUESTION NO: 23

On Cisco IOS XR software, which set of commands enables OSPF area 0 on the Gi0/0/0/0 interface that has an IPv4 IP address of 192.168.1.1/24?

- A. router ospf 1
network 192.168.1.1 0.0.0.0 area 0
- B. router ospf 1
network 192.168.1.1 255.255.255.255 area 0
- C. router ospf 1
area 0
interface gi0/0/0/0
- D. router ospfv3 1
network 192.168.1.1 0.0.0.0 area 0

E. router ospfv3 1
network 192.168.1.1 255.255.255.255 area 0

F. router ospfv3 1
area 0
network 192.168.1.0

G. router ospfv3 1
area 0
interface gi0/0/0/0

Answer: C

Explanation:

QUESTION NO: 24

Which three statements about OSPFv2 or OSPFv3 authentication are true? (Choose three.)

- A.** On Cisco IOS XR platforms, OSPFv3 authentication can be configured at the OSPF routing process, area, or interface level.
- B.** OSPF authentication on Cisco IOS and IOS XE platforms can be configured per area or per interface.
- C.** On Cisco IOS and IOS XE and IOS XR platforms, the authentication key can only be configured per interface.
- D.** OSPFv3 uses IPsec for authentication and encryption.
- E.** On Cisco IOS XR platforms, the OSPF authentication that is configured for the area overrides the authentication that is configured for the interface.
- F.** On Cisco IOS and IOS XE platforms, the two OSPFv2 authentication methods that are supported are MD5 and SHA1.

Answer: A,B,D

Explanation:

QUESTION NO: 25

Which statement is true about enabling IS-IS routing and using the net command to configure the NSAP address?

- A.** NSAP addresses have a fixed size of 20 bytes.
- B.** The NSEL must always be 00 to identify the router.
- C.** The area address follows the system ID field before the NSEL field.
- D.** An AFI of 39 indicates that the AFI is a locally administered private address.

Answer: B

Explanation:

QUESTION NO: 26

You have just enabled IS-IS on the lab testing network. You notice that IS-IS is not using the optimal path selection based on the interface bandwidth. Currently, the IS-IS routing process seems to be selecting the best path based on the hop count.

What configuration change can you make to cause IS-IS to select the optimal path based on the interface bandwidth?

- A. Enable a wide-style IS-IS metric.
- B. Enable a narrow-style IS-IS metric.
- C. Change the default auto-cost reference bandwidth to a higher value.
- D. Change the default IS-IS metric on each of the router interfaces to better correspond to the interface bandwidth.
- E. Change the interface bandwidth by using the bandwidth interface configuration mode command.

Answer: D

Explanation:

QUESTION NO: 27

You have installed a new router and configured OSPF on it. However, this new router is not able to establish an OSPF neighbor relationship with the neighbor OSPF router. Which four conditions could cause this problem? (Choose four.)

- A. mismatched OSPF hello/dead interval between the new router and the neighbor router
- B. mismatched OSPF area ID between the new router and the neighbor router
- C. mismatched OSPF authentication data between the new router and the neighbor router
- D. mismatched OSPF router ID between the new router and the neighbor router
- E. mismatched interface MTU between the new router and the neighbor router
- F. mismatched interface bandwidth between the new router and the neighbor router

Answer: A,B,C,E

Explanation:

QUESTION NO: 28

Refer to the Cisco IOS XR configuration exhibit.

```
!  
router isis 1  
net 49.0001.0100.0200.1001.00  
address-family ipv4 unicast  
metric-style wide  
address-family ipv6 unicast  
metric-style wide  
!
```

Which statement about the configuration is true?

- A. This is a Level-2-only router.
- B. This is a Level-1-only router.
- C. The maximum IS-IS path metric is 63.
- D. This router belongs to area ID 0001.
- E. This configuration is missing the net configuration for the IPv6 unicast address family.

Answer: D

Explanation:

QUESTION NO: 29

Which statement about the benefits of MPLS PHP is true?

- A. PHP is implemented to increase the performance on the egress edge LSR. PHP allows the egress edge LSR to perform only one lookup to route the IP packet based on destination IP address and routing table.
- B. PHP is implemented to increase the performance on the egress edge LSR. PHP enables the egress edge LSR to perform only one lookup based on the ingress label.
- C. PHP is implemented to increase the performance on the egress edge LSR. PHP enables the egress edge LSR to perform two lookups, first lookup based on the ingress label and a second route lookup based on the destination IP address.
- D. PHP is implemented to increase the performance on the LSR (P routers). PHP enables the LSR (P routers) to perform only one lookup based on the ingress label.
- E. PHP is implemented to increase the performance on the LSR (P routers). PHP enables the LSR

(P routers) to perform only one route lookup based on the destination IP address.

F. PHP is implemented to increase the performance on the LSR (P routers). PHP enables the LSR (P routers) to perform two lookups, first lookup based on the ingress label and a second route lookup based on the destination IP address.

G. PHP is implemented to increase the performance on the ingress edge LSR. PHP enables the ingress edge LSR to perform only one lookup based on the ingress label.

H. PHP is implemented to increase the performance on the ingress edge LSR. PHP enables the ingress edge LSR to perform two lookups, first lookup based on the ingress label and a second route lookup based on the destination IP address.

Answer: A

Explanation:

QUESTION NO: 30

Refer to the exhibit.

Media						
RP/0/RSP0/CPU0:P1# show mpls ldp forwarding						
Prefix	Label In	Label Out	Outgoing Interface	Next Hop	GR	Stale
192.168.102.0/24	16003	ImpNull	Gi0/0/0/1	192.168.112.40	N	N
10.2.10.1/32	16010	22	Gi0/0/0/1	192.168.112.40	N	N
10.10.10.100/32	16006	Unlabelled	None	10.10.10.1	N	N
<output omitted>						

Which three statements about MPLS LDP operations are true?

A. The incoming label for 10.2.10.1/32 is 16010 (which is allocated by the local router), and the outgoing label is 22 (as advertised by the next-hop router).

B. The 16003 and 16010 labels are learned from the 192.168.112.40 LDP neighbor.

C. The ImpNull outgoing label for the 192.168.102.0/24 network means that the outgoing label should be removed when sending packets to the 192.168.112.40 next-hop router, when the incoming label is 16003.

D. The ImpNull outgoing label for the 192.168.102.0/24 network means that the local router has not received a label for that network from its neighbor.

E. The Unlabeled outgoing label for 10.10.10.100/32 indicates that a label pop is required when sending packets to the 10.10.10.1 next-hop router.

F. The outgoing label of 22 for 10.2.10.1/32 is learned from the 192.168.112.40 LDP neighbor.

Answer: A,C,F

Explanation:

QUESTION NO: 31

On Cisco IOS XR software, how is LDP enabled on an interface?

- A. LDP is automatically enabled globally on all interfaces, when ip cef is enabled globally.
- B. LDP is enabled on each interface, using the mpls ip interface configuration command.
- C. LDP is enabled on each interface, using the interface command under mpls ldp (MPLS LDP configuration mode).
- D. LDP is enabled globally on all interfaces, using the mpls ldp global configuration command.
- E. LDP is enabled globally on all interfaces, using the mpls ip global configuration command.

Answer: C

Explanation:

QUESTION NO: 32

Which four statements about REP configurations on a Cisco ME 3400 switch port are true? (Choose four.)

- A. The port must be an NNI type and must be in trunk mode.
- B. Use the rep segment number command to enable REP on the switch port.
- C. A REP segment cannot be wrapped into a ring topology. REP supports only a topology where each REP segment has two exit points via two edge switches.
- D. The port where the segment terminates is called the edge port.
- E. With REP, at least one port is always blocked in any given segment-that is, the alternate port.
- F. If a failure occurs within the segment, the blocked port goes to the forwarding state after it goes through the listening state and the learning state.

Answer: A,B,D,E

Explanation:

QUESTION NO: 33

A VLAN is a logical grouping of switch ports that belong to which two of these? (Choose two.)

- A. the same virtual switch cluster

- B. the same IP subnet
- C. the same collision domain
- D. the same broadcast domain
- E. the same virtual routing and forwarding instance

Answer: B,D

Explanation:

QUESTION NO: 34

Refer to the exhibit.

```
SwitchX#configure terminal
SwitchX(config)#interface fa0/11
SwitchX(config-if)#switchport mode trunk
SwitchX(config-if)#end
```

Which statement is true?

- A. VLAN 1 traffic is sent untagged on the fa0/11 trunk port.
- B. All VLANs traffic, including the native VLAN traffic, is tagged with a VLAN ID when it is sent over the fa0/11 trunk port.
- C. The switchport trunk allowed vlan command is not configured; therefore, no VLANs are allowed on the trunk.
- D. The encapsulation type that is used is 802.1ad.
- E. The switchport trunk native vlan command is not configured; therefore, the trunk is not operational.

Answer: A

Explanation:

QUESTION NO: 35

Refer to the partial configurations exhibit.

```
! ME3400
!  
ip routing
!  
vlan 10
vlan 20
interface Fa0/1
switchport access vlan 10
interface Fa0/2
switchport access vlan 20
!
```

What additional configurations are required to enable inter-VLAN routing for VLANs 10 and 20 on the Cisco ME 3400 switch using the metro IP access image?

A. interface Fa0/1

```
ip address 192.168.10.1 255.255.255.0
```

```
!
```

```
interface Fa0/2
```

```
ip address 192.168.20.1 255.255.255.0
```

```
!
```

B. interface Fa0/1

```
ip address 192.168.10.1 255.255.255.0
```

```
!
```

```
interface Fa0/2
```

```
ip address 192.168.20.1 255.255.255.0
```

```
!
```

```
router eigrp 1
```

```
network 192.168.10.0
```

```
network 192.168.20.0
```

```
!
```

C. interface vlan 10

```
ip address 192.168.10.1 255.255.255.0
```

```
!
```

```
interface vlan 20
```

```
ip address 192.168.20.1 255.255.255.0
```

```
!
```

D. interface Fa0/1

```
encapsulation dot1q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
!
```

```
interface Fa0/2
```

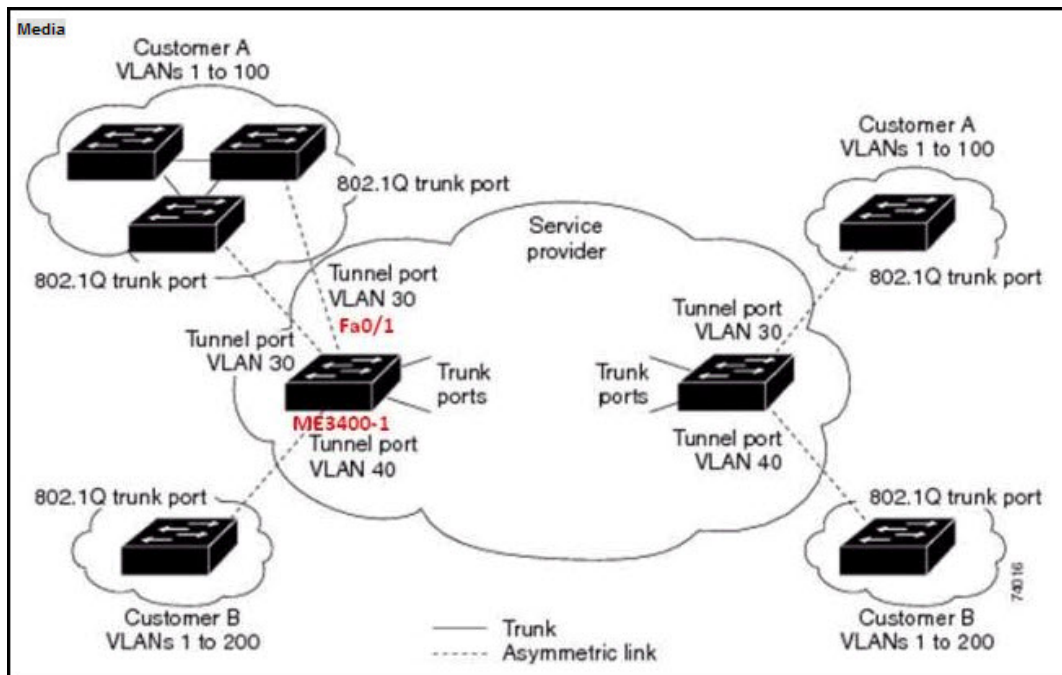
```
encapsulation dot1q 20
```

```
ip address 192.168.20.1 255.255.255.0
```

!

Answer: C**Explanation:****QUESTION NO: 36**

Refer to the topology diagram in the exhibit.



Which two interface configuration commands on the Cisco ME 3400-1 switch Fa 0/1 interface are required to support the VLANs for Customer A to be trunked across the service provider network? (Choose two.)

- A. switchport access vlan 30
- B. switchport mode trunk
- C. switchport mode dot1q-tunnel
- D. switchport trunk allowed vlan 1-100

Answer: A,C**Explanation:****QUESTION NO: 37**

Which IPv6 address block is reserved for 6to4 tunneling?

- A. 2000::/16
- B. 2001::/16
- C. 2002::/16
- D. 3ffe::/16
- E. fe80::/16

Answer: C

Explanation:

QUESTION NO: 38

When upgrading a Cisco ASR 1001 Router, when is the request platform software package expand file bootflash:image-name command required?

- A. to copy the consolidated Cisco IOS XE image to the bootflash:
- B. to perform Cisco IOS ISSU on the standby route processor
- C. to run the router using a consolidated package
- D. to run the router using individual subpackages

Answer: D

Explanation:

QUESTION NO: 39

Which Cisco router platform supports running either the Cisco IOS or IOS XR operating system?

- A. Cisco CRS
- B. Cisco ASR9k
- C. Cisco ASR1K
- D. Cisco ISR-G2
- E. Cisco 12000 GSR

Answer: E

Explanation:

QUESTION NO: 40

You want to configure HSRP between a Cisco IOS and a Cisco IOS XR router for the 192.0.2.0/24 subnet. Half of the PCs on the 192.0.2.0/24 subnet are configured to use 192.0.2.1 as the default gateway, and the other half of the PCs are configured to use 192.0.2.254 as the default gateway. The intent is to load balance the traffic across both routers.

Which two IOS and IOS XR configurations are needed? (Choose two.)

A. ! IOS-XR

```
interface GigabitEthernet 0/0/0/0
ip address 192.0.2.3 255.255.255.0
router hsrp
interface GigabitEthernet 0/0/0/0
hsrp 1 ipv4 192.0.2.1
hsrp 1 priority 95
hsrp 1 preempt
hsrp 2 ipv4 192.0.2.254
hsrp 2 priority 105
hsrp 2 preempt
```

B. ! IOS-XR

```
interface GigabitEthernet 0/0/0/0
ip address 192.0.2.1 255.255.255.0
router hsrp
interface GigabitEthernet 0/0/0/0
hsrp 1 ipv4 192.0.2.1
hsrp 1 priority 99
hsrp 1 preempt
hsrp 2 ipv4 192.0.2.254
hsrp 2 priority 101
hsrp 2 preempt
```

C. ! IOS-XR

```
interface GigabitEthernet 0/0/0/0
ip address 192.0.2.3 255.255.255.0
router hsrp
interface GigabitEthernet 0/0/0/0
hsrp 1 ipv4 192.0.2.1
hsrp 1 priority 1
hsrp 1 preempt
hsrp 2 ipv4 192.0.2.254
hsrp 2 priority 1
hsrp 2 preempt
```

D. ! IOS

```
interface GigabitEthernet 0/0
ip address 192.0.2.2 255.255.255.0
standby 1 ip 192.0.2.1
```

```
standby 1 priority 105
standby 1 preempt
standby 2 ip 192.0.2.254
standby 2 priority 95
standby 2 preempt
E. ! IOS
interface GigabitEthernet 0/0
ip address 192.0.2.254 255.255.255.0
standby 1 ip 192.0.2.1
standby 1 preempt
standby 2 ip 192.0.2.254
standby 2 preempt
F. ! IOS
interface GigabitEthernet 0/0
ip address 192.0.2.2 255.255.255.0
standby 1 ip 192.0.2.1
standby 1 preempt
standby 1 priority 2
standby 2 ip 192.0.2.254
standby 2 preempt
standby 2 priority 2
```

Answer: A,D

Explanation:

QUESTION NO: 41

On Cisco IOS XR software, which two of the address-family command options in IS-IS configuration mode are valid? (Choose two)

- A. address-family clns
- B. address-family ipv4 unicast
- C. address-family ipv6 unicast
- D. address-family vpnv4
- E. address-family vpnv6

Answer: B,C

Explanation:

QUESTION NO: 42

On Cisco routers, the address-family configuration command in BGP configuration mode is used to enable which BGP feature?

- A. BGP route policy
- B. multiprotocol BGP
- C. BGP policy accounting
- D. BGP communities

Answer: B

Explanation:

QUESTION NO: 43

Your switch network has been configured to support multiple VLANs. In order for the users on one VLAN to communicate with users on another VLAN, which additional configuration is required?

- A. Enable a routing protocol like EIGRP on a router or on a Layer 3 capable switch
- B. Enable inter-VLAN routing on a router or on a Layer 3 capable switch
- C. Enable a switch virtual interface on the router, using the interface vlan global configuration command
- D. Enable a switch virtual interface on the switch, using the encapsulation dot1q
- E. subinterface configuration command
- F. Enable IP routing on a Layer 2 switch

Answer: B

Explanation:

QUESTION NO: 44

Within an MPLS domain, which table is used by the label switch routers to make forwarding decisions when a labeled packet is received?

- A. FIB
- B. RIB
- C. LFIB
- D. LIB
- E. CEF

Answer: C

Explanation:

QUESTION NO: 45

On Cisco IOS XR software, what is the default commit option?

- A. best-effort
- B. psuedo-atomic
- C. replace
- D. force

Answer: B

Explanation:

QUESTION NO: 46

Refer to the show command exhibit.

```
RP/0/RSP0/CPU0:PE7(config)#show configuration
Building configuration...
!! IOS XR Configuration 4.1.0
hostname PE7
username execuser
  group operator
  password 7 02050D480809
!
cdp
line console
  exec-timeout 10 0
!
interface Loopback0
  ipv4 address 10.7.1.1 255.255.255.255
  ipv6 address 2001:db8:10:7:1::1/128
  ipv6 enable
!
interface MgmtEth0/RSP0/CPU0/0
  cdp
  ipv4 address 10.10.10.40 255.255.255.0
!
interface MgmtEth0/RSP0/CPU0/1
  shutdown
!
interface GigabitEthernet0/0/0/0
  cdp
  ipv4 address 192.168.107.70 255.255.255.0
  ipv6 address 2001:db8:192:168:107::70/64
  ipv6 enable
  speed 100
  duplex full
!
!
route-policy pass
  pass
end-policy
!
end
!
```

Which statement is true?

- A. It shows the result of the target configuration merged with the running configuration.
- B. It shows the contents of the uncommitted configuration.

- C. It shows the contents of the failed configuration.
- D. It shows the contents of the running configuration.

Answer: B

Explanation:

QUESTION NO: 47

When configuring a new VLAN on Cisco IOS switches, which configuration parameter is required?

- A. VLAN name
- B. native VLAN ID
- C. VLAN ID
- D. VLAN map
- E. VLAN MTU

Answer: C

Explanation:

QUESTION NO: 48

What is a requirement for performing Cisco IOS ISSU on the Cisco IOS XE-based ASR 1006 and ASR 1013 routers?

- A. must use consolidated package mode
- B. must use subpackage mode
- C. must have redundant RPs
- D. must have owner and nonowner SDRs

Answer: C

Explanation:

QUESTION NO: 49

Refer to the partial Cisco IOS router configuration exhibit.

```
interface FastEthernet0/0
no ip address
no shut
!
interface FastEthernet0/0.1
ip address 192.168.1.1 255.255.255.0
!
interface FastEthernet0/0.2
ip address 192.168.2.1 255.255.255.0
!
interface FastEthernet0/0.3
ip address 192.168.3.1 255.255.255.0
!
```

Which statement is true?

- A.** To support the subinterface configuration, the Fa0/0 main interface configuration is missing the encapsulation dot1Q command.
- B.** The Fa0/0 interface/subinterfaces are configured correctly to operate as a trunk port to provide inter-VLAN routing for three VLANs (VLANs 1, 2, and 3).
- C.** To provide inter-VLAN routing, the Fa0/0.1, Fa0/0.2 and Fa0/0.3 subinterface configuration is missing the encapsulation dot1Q vlan-id subinterface configuration command.
- D.** To provide inter-VLAN routing, the router is missing the vlan 1, vlan 2, and vlan 3 global configuration commands.

Answer: C

Explanation:

QUESTION NO: 50

Refer to the partial Cisco IOS XR BGP configuration exhibit.

```
router bgp 64500
 address-family ipv4 unicast
  network 10.1.1.1/32
 !
 address-family ipv6 unicast
  network 2001:db8:10:1:1::1/128
 !
 neighbor 10.2.1.1
  remote-as 64500
  update-source Loopback0
  address-family ipv4 unicast
  !
 !
 neighbor 192.168.101.11
  remote-as 64501
  password encrypted 13061E010803
  address-family ipv4 unicast
  route-policy Test1 in
  route-policy Test1 out
  !
 !
 neighbor 2001:db8:10:2:1::1
  remote-as 64500
  address-family ipv6 unicast
  !
 !
 neighbor 2001:db8:192:168:101::11
  remote-as 64501
  address-family ipv6 unicast
  route-policy Test1 in
  route-policy Test1 out
  !
 !
 !
end
```

Which two statements are true? (Choose two)

- A. This router uses its Loopback 0 interface IP address when establishing BGP peering with the 10.2.1.1 router.
- B. This router uses its Loopback 0 interface IP address when establishing BGP peering with the 192.168.101.11 router.
- C. Test1 refers to a route policy that is defined using the RPL.
- D. Both the IBGP and EBGP sessions will be authenticated.

E. The IBGP session is missing the mandatory route-policy configuration.

Answer: A,C

Explanation:

QUESTION NO: 51

When is a route-policy configuration needed on a Cisco IOS XR router using RPL?

- A. when enabling an IGP routing protocol configuration
- B. when enabling an EBGP peering configuration
- C. when enabling an IBGP peering configuration
- D. when enabling route redistribution
- E. when enabling an LDP peering configuration

Answer: B

Explanation:

QUESTION NO: 52

Refer to the partial Cisco IOS router configuration exhibit.

```
interface Tunnel0
ipv6 address 2001:db8:3::2/64
tunnel source GigabitEthernet0/0
tunnel destination 209.165.201.1
tunnel mode ipv6ip
!
```

Which type of tunnel configuration is shown?

- A. GRE tunnel
- B. manual 6in4 tunnel
- C. 6to4 automatic tunnel
- D. 6RD
- E. IPv4-in-IPv6 tunnel

Answer: B

Explanation:

QUESTION NO: 53

Which two protocols are most often deployed in Cisco IP NGN multiservices core networks to support services like VPNs? (Choose two.)

- A. MPLS
- B. GRE
- C. mGRE
- D. SSL
- E. VTI
- F. multiprotocol BGP

Answer: A,F

Explanation:

QUESTION NO: 54

Which segment protocol provides fast and predictable convergence (typically within 50 ms) in Layer 2 Ethernet ring topologies?

- A. Spanning Tree Protocol
- B. Flex Links
- C. Resilient Ethernet Protocol
- D. Spatial Reuse Protocol
- E. Resilient Packet Ring

Answer: C

Explanation:

QUESTION NO: 55

Within an MPLS domain, which table is used by the ingress edge LSR to make forwarding decisions when an unlabeled IP packet is received?

- A. FIB
- B. LFIB

- C. LIB
- D. MP-BGP
- E. VRF

Answer: A

Explanation:

QUESTION NO: 56

Which three statements about access control lists on a Cisco IOS router are true? (Choose three)

- A. The more specific ACL entries should be placed at the top of the ACL.
- B. The generic ACL entries should be placed at the top of the ACL, to filter general traffic and reduce noise on the network.
- C. ACLs always search for the most specific entry before taking any filtering action.
- D. Router-generated packets cannot be filtered by the interface ACLs on the router.
- E. Extended ACLs should be placed as close to the destination as possible.
- F. There must be at least one permit statement in an ACL, or all traffic is denied.

Answer: A,D,F

Explanation:

QUESTION NO: 57

Which Cisco IOS access list permits HTTP traffic that is sourced from host 10.1.129.100 port 3030 and that is destined to host 192.168.1.10?

- A. access-list 101 permit tcp any eq 3030
- B. access-list 101 permit tcp 10.1.128.0 0.0.1.255 eq 3030 192.168.1.0 0.0.0.15 eq www
- C. access-list 101 permit tcp 10.1.129.0 0.0.0.255 eq www 192.168.1.10 0.0.0.0 eq www
- D. access-list 101 permit tcp host 192.168.1.10 eq 80 10.1.0.0 0.0.255.255 eq 3030
- E. access-list 101 permit tcp 192.168.1.10 0.0.0.0 eq 80 10.1.0.0 0.0.255.255
- F. access-list 101 permit ip host 10.1.129.100 eq 3030 host 192.168.1.100 eq 80

Answer: B

Explanation:

QUESTION NO: 58

What is dual IOS mode on the Cisco ASR 1001 Router?

- A. redundant IOS processes that are running on the active RP and standby RP
- B. active and standby IOS processes that are running on a single RP
- C. separate Cisco IOS XE and IOS XR processes that are running on a single RP
- D. separate Cisco IOS XE and IOS XR processes that are running on two different RPs
- E. checkpointed redundant IOS processes that are running on two different RPs in active/active mode

Answer: B

Explanation:

QUESTION NO: 59

Which Cisco platform runs the Cisco IOS XE operating system?

- A. Cisco ASR9K
- B. Cisco CRS
- C. Cisco 12000 GSR
- D. Cisco ASR1K
- E. Cisco ME 3400
- F. Cisco ISR-G2

Answer: D

Explanation:

QUESTION NO: 60

A service provider needs to implement a managed CE device that supports Cisco IOS ISSU, and the current CE support staff is highly trained in Cisco IOS only. Which Cisco router platform should the service provider implement as the managed CE device?

- A. Cisco ASR1K
- B. Cisco ISR-G2
- C. Cisco ASR9K
- D. Cisco 12000 GSR
- E. Cisco 7200
- F. Cisco 7600

Answer: A

Explanation:

QUESTION NO: 61

On Cisco IOS XR software, what are the four valid address-family configuration command options in static route configuration mode? (Choose four.)

- A. address-family ipv4 unicast
- B. address-family ipv6 unicast
- C. address-family ipv4 multicast
- D. address-family ipv6 multicast
- E. address-family vpnv4
- F. address-family vpnv6

Answer: A,B,C,D

Explanation:

QUESTION NO: 62

Which three statements about service provider network requirements are true? (Choose three.)

- A. Service provider networks are becoming more specialized, with providers offering only one service.
- B. Service providers connect customers to the Internet.
- C. Jitter and packet loss are no longer an issue.
- D. Multiple access technologies are supported.
- E. Customers can connect directly using copper, wireless, or fiber-optic links.

Answer: B,D,E

Explanation:

QUESTION NO: 63

Which three are common ISP access network technologies? (Choose three.)

- A. DSL
- B. Cable modem
- C. DWDM

- D. MPLS
- E. PON
- F. MP-BGP

Answer: A,B,E

Explanation:

QUESTION NO: 64

Which three Cisco platforms are classified as core routers? (Choose three.)

- A. Cisco XR 12000
- B. Cisco ME 3400
- C. Cisco ASR 1006
- D. Cisco CRS-3
- E. Cisco ASR 9010
- F. Cisco 3900

Answer: A,D,E

Explanation:

QUESTION NO: 65

Which statement about Cisco IOS XE software is true?

- A. It uses a two-stage configuration process.
- B. It is identical in look and feel to Cisco IOS software.
- C. It runs on the Cisco ASR 9010 platform.
- D. Packages can be added using the install add and install activate commands.
- E. It runs on the Cisco ME 3400 platform.

Answer: B

Explanation:

QUESTION NO: 66

Which three statements about VLAN implementation in a Layer 2 switch environment are true? (Choose three.)

- A. The Cisco IOS switchport access vlan interface configuration command is used to configure the native VLAN on the switch port.
- B. 802.1AD supports VLAN overlap.
- C. VLANs can span across multiple switches.
- D. 802.1Q encapsulation is used to identify different VLANs on a trunk port.
- E. Inter-VLAN routing is performed between VLAN tunnel ports.

Answer: B,C,D

Explanation:

QUESTION NO: 67

Refer to the configuration example exhibit.

```
S1(config)# interface f0/1
S1(config-if)# switchport access vlan 118
S1(config-if)# switchport mode dot1q-tunnel
S1(config-if)# interface f0/3
S1(config-if)# switchport access vlan 209
S1(config-if)# switchport mode dot1q-tunnel
```

Which statement is true?

- A. A trunk port has been created and VLAN has been allowed.
- B. An access port has been created on a UNI.
- C. A customer VLAN of 209 is configured.
- D. A QinQ VLAN of 118 is configured.

Answer: D

Explanation:

QUESTION NO: 68

Which two statements about the link state routing process are true? (Choose two.)

- A. It uses the DUAL algorithm.
- B. It uses metrics such as AS path.
- C. All routers in the area have link state databases.

- D. The administrative distance is 1 by default.
- E. Each router in the area floods LSPs to all neighbors.

Answer: C,E

Explanation:

QUESTION NO: 69

Which two statements about carrier-grade NAT are true? (Choose two.)

- A. It conserves IPv4 addresses.
- B. A service provider issues private IP addresses to its customers.
- C. A service provider translates the private IP address of its customer to another private IP address.
- D. It is implemented on the CE and PE routers.
- E. It is designed to simplify IPv6 addressing.

Answer: A,B